

PHOSPHATES

When metals combine with the phosphate radical (PO_4), phosphate minerals are produced. These minerals form a large group of over 200 – however, many are very rare. Some minerals in this group are brightly coloured. They usually form by the alteration of sulfide minerals, but they are sometimes primary minerals. A number of phosphates are radioactive.

HYDROXYLHERDERITE

Monoclinic • H 5–5½ • SG 2.95

Hydroxylherderite is a calcium beryllium phosphate. It occurs as pale yellow or greenish crystals, with a glassy lustre, in granitic pegmatites.



DUFRÉNITE

Monoclinic

H 3½–4½ • SG 3.10–3.34

This hydrated phosphate of iron and calcium mainly occurs as green to black masses or crusts in altered veins and iron ores.



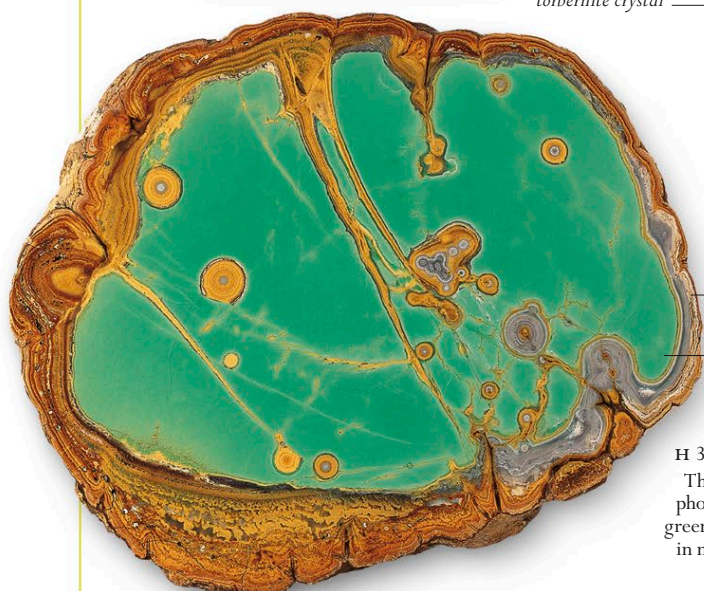
AUTUNITE

Orthorhombic

H 2–2½ • SG 3.05–3.2

Radioactive autunite is a lemon-yellow or pale green hydrated phosphate of calcium and uranium. It occurs where uranium minerals are altered.

tabular
torbernite crystal



VARISCITE

Orthorhombic

H 3½–4½ • SG 2.56–2.61

This hydrated aluminium phosphate usually occurs as green microcrystalline masses in nodules, veins, or crusts.



TURQUOISE

Triclinic • H 5–6 • SG 2.6–2.8

A sought after gem for thousands of years, this hydrated phosphate of copper and aluminium is found in altered igneous rocks.



PYROMORPHITE

Hexagonal • H 3½–4 • SG 7.04

A lead phosphate chloride, variably coloured greenish, orange, yellowish or brownish, pyromorphite forms in the oxidized zones of lead deposits.

prismatic
apatite crystal



APATITE

Hexagonal

H 5 • SG 3.16–3.22

Apatite is the group name used for three similar calcium phosphate minerals: fluorapatite, chlorapatite, and hydroxylapatite.



TORBERNITE

Tetragonal • H 2–2½ • SG 3.22

This copper uranium phosphate hydrate is related to autunite and occurs in similar geological environments. It is a radioactive mineral.

sliced nodule

waxy lustre



TRIPLITE

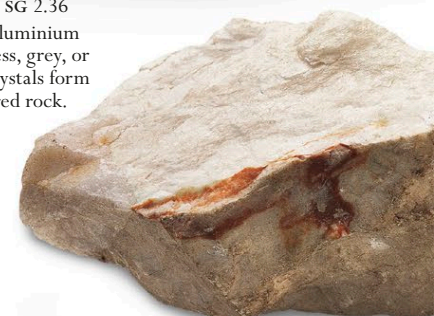
Monoclinic • H 5–5½ • SG 3.5–3.9

Triplite is a phosphate of manganese and fluorine, sometimes with iron. It forms in granitic pegmatites.

WAVELLITE

Orthorhombic • H 3½–4 • SG 2.36

Wavellite is a rare hydrated aluminium phosphate hydroxide. Colourless, grey, or greenish, glassy, needle-like crystals form radiating aggregates in altered rock.



AMBLYGONITE

Triclinic • H 5½–6 • SG 3.04–3.11

This rare lithium aluminium fluorophosphate mainly forms as masses, but crystals occur in Zimbabwe and Brazil.

